UKAPA7 Layer U Partial B-Side Offset Report

This report repeats the UKAPA7 comparison workflow for the U layer using:

- UK factory measurements in `UKAPA7U.json`
- Chicago factory measurements in `data/tension_summaries/tension_summary_UKAPA7_U.csv`

Residual definition throughout: `Chicago - UK`.

Data Availability

- The UK JSON contains both side A and side B measurements.
- The Chicago summary CSV contains only side B data.
- The available Chicago side-B wires span wire numbers `241` through `960`.
- The current CSV contains `621` non-null side-B measurements and `0` non-null side-A measurements.

Because there is no Chicago side-A data in the summary CSV, this report is necessarily limited to the available side-B subset.

Working Assumption

For this U-layer analysis, the working assumption is:

- there is no side-B reversal
- there may be a constant wire-number offset

So the tested model family is a simple shift:

`Chicago B wire w` -> `UK B wire w + k`

with no reversal term.

Executive Summary

- Under the current indexing, the partial B-side comparison is already strongly negative on average, with mean residual about `-0.958 N`.
- Scanning constant shifts from `-80` to `+80` shows the best correlation at shift `-7`.
- That shift improves correlation from `0.355` to `0.425` and reduces the residual width from `0.654 N` to `0.607 N`.

- The improvement is real but not transformative. The Chicago and UK traces remain substantially offset even after the best tested small shift.

Source Artifacts

- Raw B comparison, current indexing: `data/tension_plots/tension_raw_B_baseline_UKAPA7_U.png`
- Raw B comparison, best offset: `data/tension_plots/tension_raw_B_shifted_UKAPA7_U.png`
- Residual comparison, baseline vs best offset: `data/tension_plots/tension_residual_B_offset_comparison_UKAPA7_U.png`
- Offset scan: `data/tension_plots/tension_shift_scan_B_UKAPA7_U.png`
- Aligned comparison table: `data/tension_summaries/tension_B_offset_comparison_UKAPA7_U.csv`

Raw B-Side Data, Current Indexing

![UKAPA7 U raw B baseline](../data/tension_plots/tension_raw_B_baseline_UKAPA7_U.png){ width=100% }

Current-Indexing Summary

Comparison	Count	Chicago Mean (N)	UK Mean (N)	Corr	Mean Diff (N)	Std Diff (N)	MAE (N)
Side B baseline	621	5.372	6.330	0.355	-0.958	0.654	0.998

Raw B-Side Data, Best Constant Offset

![UKAPA7 U raw B shifted](../data/tension_plots/tension_raw_B_shifted_UKAPA7_U.png){ width=100% }

Best-Offset Summary

Comparison	Count	Chicago Mean (N)	UK Mean (N)	Corr	Mean Diff (N)	Std Diff (N)	MAE (N)
Side B shift `-7`	621	5.372	6.323	0.425	-0.950	0.607	0.977

Interpretation

- A simple negative shift of `-7` wires produces the best correlation in the tested range.
- The improvement is visible in both the line plot and the histogram overlap.
- Even after the shift, the Chicago values remain systematically below the UK

values by about `0.95 N` on average.

Residual Comparison

This figure compares the residuals for:

- baseline indexing
- the best tested constant offset, shift `-7`

![UKAPA7 U residual comparison](../data/tension_plots/tension_residual_B_offset_comparison_UKAPA7_U.png){ width=100% }

Residual Statistics

Model	Count	Corr	Mean (N)	Median (N)	Std (N)	MAE (N)	RMSE (N)	Negative Fraction
Baseline	621	0.355	-0.958	-0.877	0.654	0.998	1.160	0.958
Shift `-7`	621	0.425	-0.950	-0.850	0.607	0.977	1.128	0.974

Interpretation

- The best constant shift tightens the residual distribution and increases the correlation.
- The residuals remain strongly negative even after the shift, which means the dominant effect is still a large average Chicago-versus-UK difference.
- Since only part of side B is present in the CSV, this should be interpreted as a partial-window comparison, not a statement about the entire layer.

Offset Scan

![UKAPA7 U shift scan](../data/tension_plots/tension_shift_scan_B_UKAPA7_U.png){ width=100% }

Interpretation

- The correlation curve peaks near shift `-7`.
- The residual standard deviation also reaches a local minimum around the same region.
- Larger negative shifts can lower MAE somewhat, but they do not give the best correlation and are less compelling as a simple indexing-offset explanation.

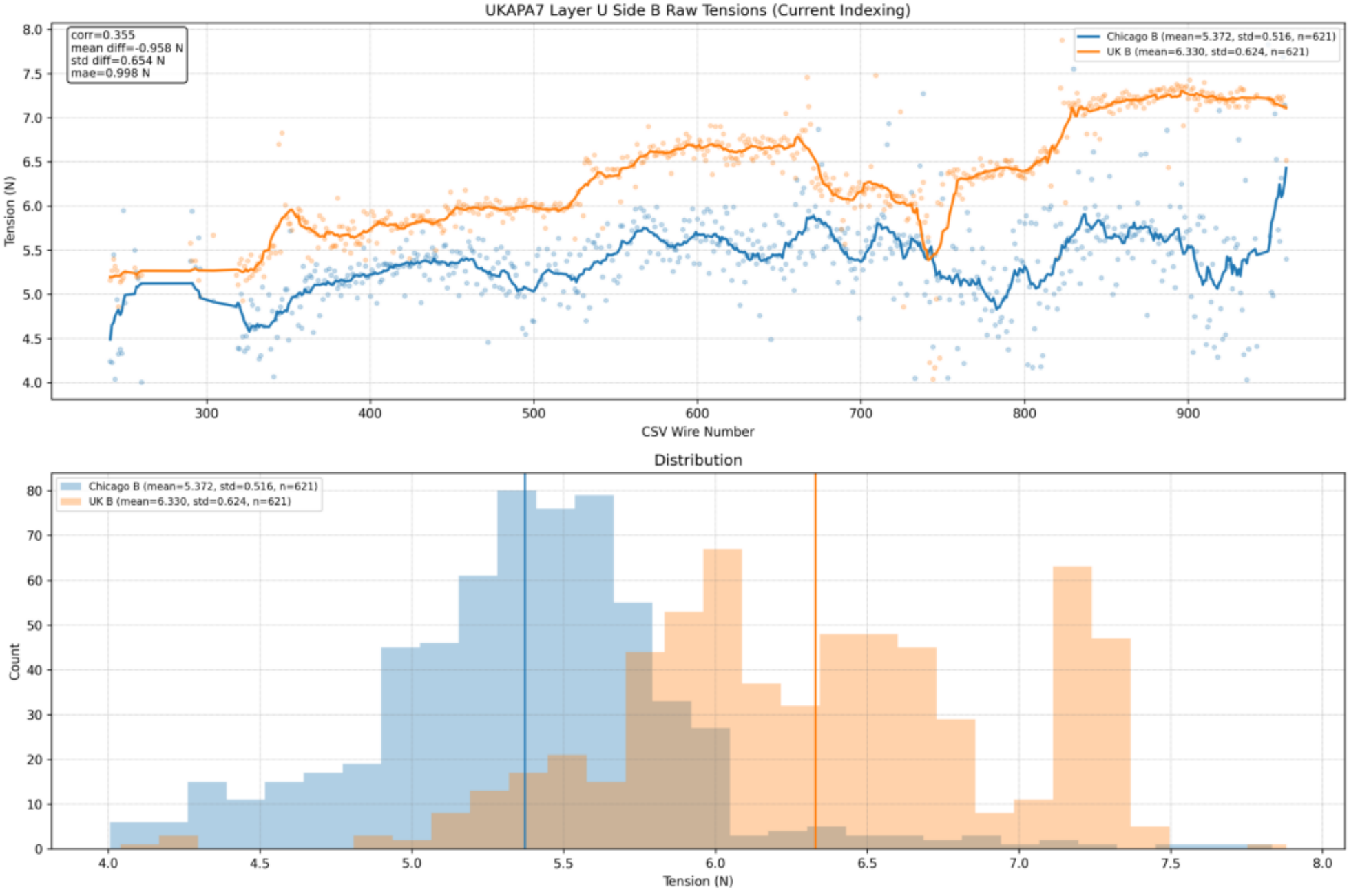
Conclusion

For the available U-layer data, there is no need to invoke a reversal model.

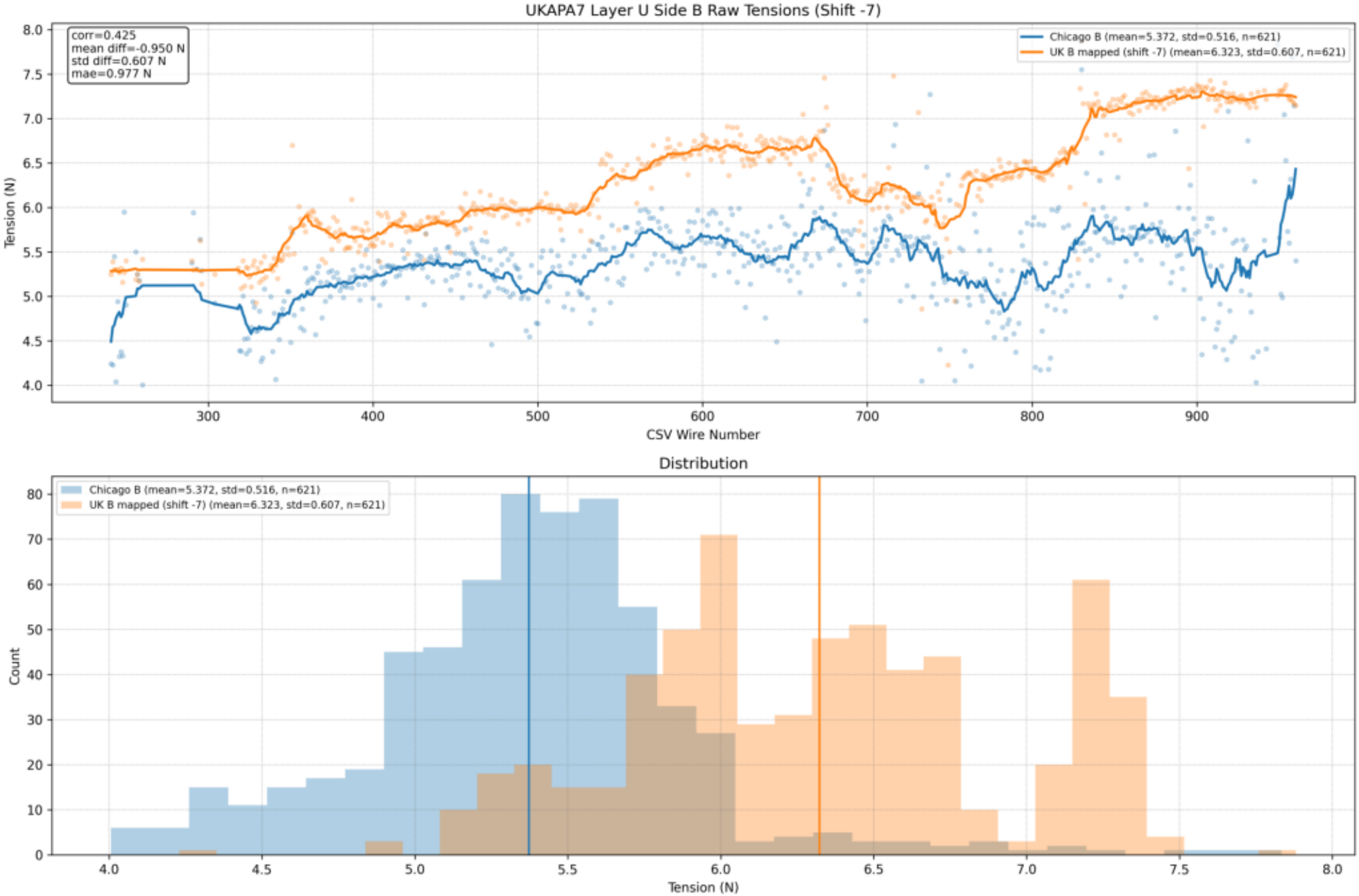
The Chicago summary contains only a partial side-B window, and within that window the best simple index-offset explanation is a shift of `-7` wires.

That shift improves the match, but only moderately. The comparison still shows an overall large negative offset between Chicago and UK measurements, and the absence of Chicago side-A data limits how far the interpretation can be pushed.

tension_raw_B_baseline_UKAPA7_U.png



tension_raw_B_shifted_UKAPA7_U.png



tension_residual_B_offset_comparison_UKAPA7_U.png

tension_shift_scan_B_UKAPA7_U.png